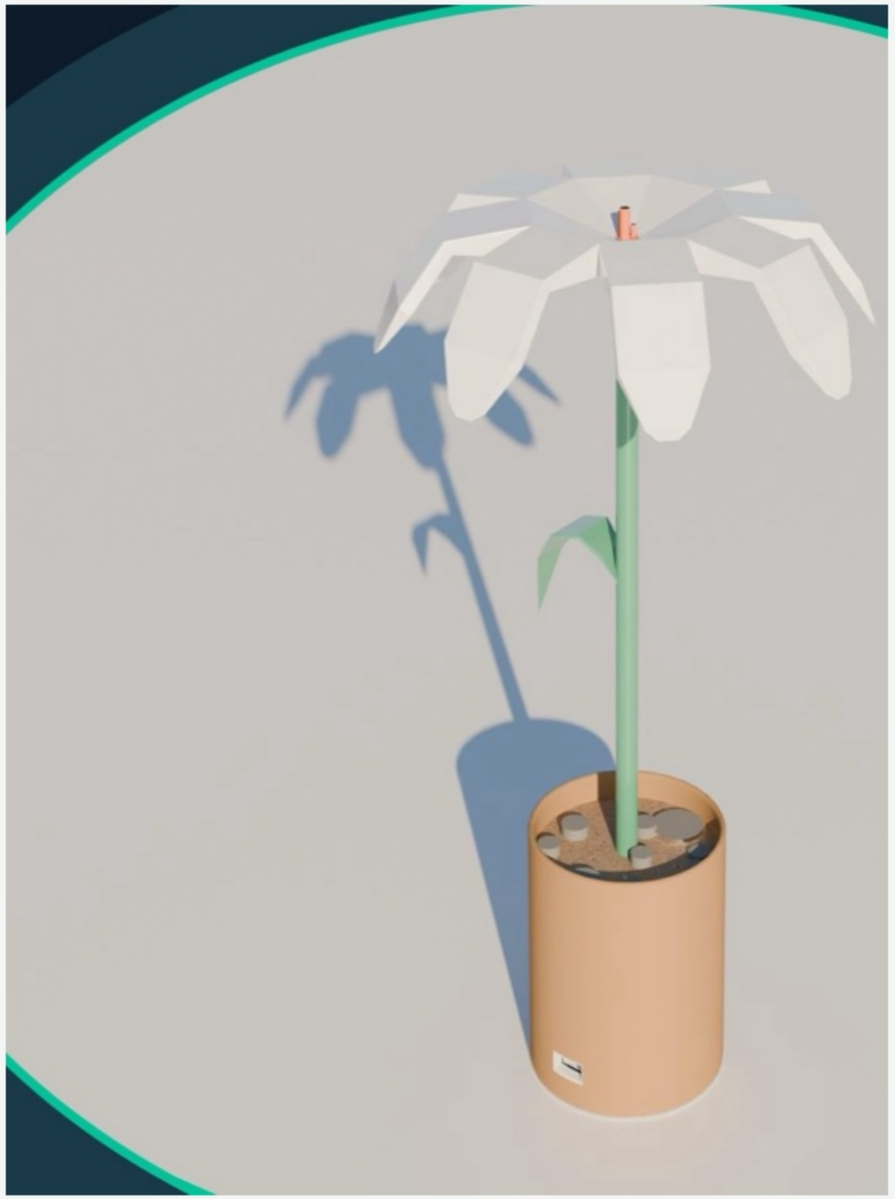


Glowing Lily, CO_2 Sensor

A botanical ambient interface for indoor cognitive health

Tasked with designing a device to translate an undetectable environmental metric into a human sense, I developed an ambient indoor CO_2 monitor that alerts studying students to cognitive fatigue before it sets in—without screens, notifications, or apps.

Discipline

Product & Interaction

Context

CSM 'Seventh Sense'

Tools

Fusion 360, SolidWorks

Outcome

A complete 1:1 physical and CAD technical package integrating a Sensirion SCD40 sensor, NeoPixel ring, and Seeeduino XIAO inside a multi-part 3D-printed housing.

The brief & territory — sense of environmental stress

The Brief — 'Seventh Sense'

Humans perceive the world through five traditional senses plus proprioception. Yet modern indoor life exposes us to invisible environmental shifts—air composition, electromagnetic fields, particulate matter—that directly impair our cognition without triggering a single human receptor.

My Territory — Indoor Air Quality & Cognitive Decline

I chose to investigate the invisible accumulation of carbon dioxide (CO_2) in small, enclosed study bedrooms and libraries. When students work with doors and windows shut, metabolic CO_2 rises from 400 ppm to over 1,500 ppm in under two hours—causing drowsiness, headaches, and a 15–50% drop in decision-making performance.

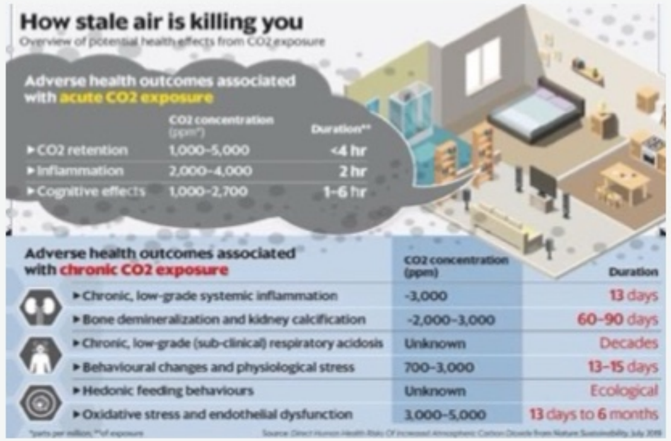
01 **Invisible Stressor** — Odourless and colourless: humans cannot detect rising CO_2 until headache and lethargy set in.

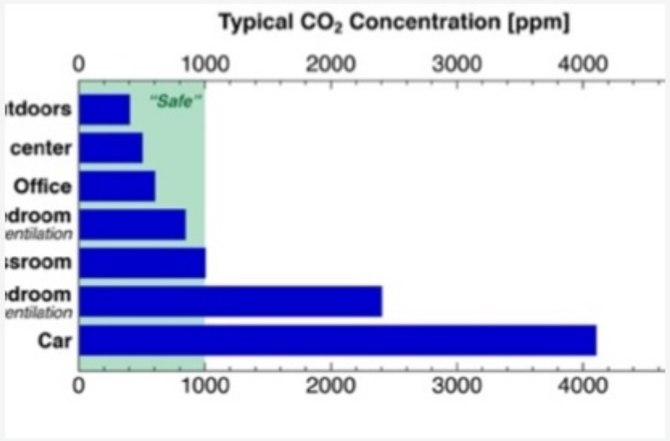
02 **Misattributed Fatigue** — Students consistently blame boredom, screen fatigue, or lack of discipline rather than stale air.

03 **Zero-Screen Mandate** — A productivity tool must not add another app notification or digital readout to a crowded desk.

04 **Peripheral Calm** — Inspired by calm technology and biophilic design—communicating state shifts through natural light.

Secondary research — the indoor CO_2 problem







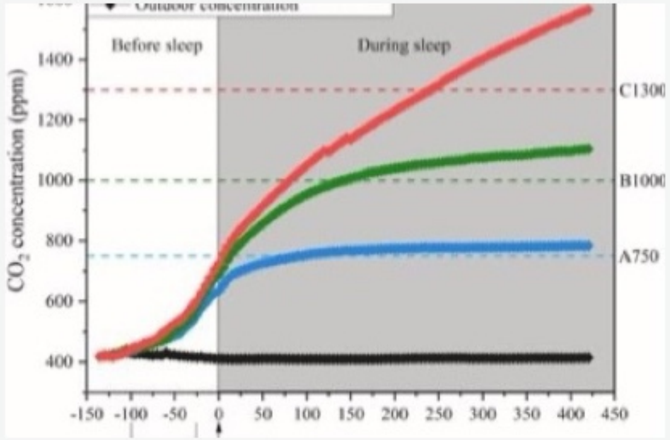


Fig. 01–04 — Secondary literature & architectural air-quality studies: enclosed study rooms exceed 1,000 ppm within 45–90 minutes

Harvard T.H. Chan School of Public Health Findings

Controlled office and classroom studies demonstrate that at 1,000 ppm CO_2, human cognitive scores drop by 15%, and at 1,400 ppm complex strategic thinking drops by 50%.

Zone	CO_2 Level	Cognitive & Physiological Impact
Fresh	**400–800 ppm**	Optimal indoor air quality; peak cognitive focus
Stale	**800–1000 ppm**	Air quality falling; subtle onset of drowsiness
Alert	**1000–1500 ppm**	15% drop in cognitive scores; ventilate room now
Severe	**1500–2500 ppm**	50% drop in complex thinking; headache & lethargy

Primary research — productivity crash survey





The Attribution Blind Spot

I surveyed students working in enclosed bedrooms to measure how long they studied before experiencing a cognitive crash—and what they blamed for the slump.

01 **Person A — Crashed at 1.5 hrs** — Blamed 'Too much time spent on the same thing' (Behavioural attribution).

02 **Person B — Crashed at 2.0 hrs** — Blamed 'Lack of stimulation and boredom' (Behavioural attribution).

03 **Person C — Crashed at 2.0 hrs** — Blamed 'Doing one activity for too long' — 0 of 3 recognised CO_2 buildup.

Fig. 05–06 — Primary survey & study desk documentation: 100% of respondents crashed within 1.5–2 hours in closed rooms

Ethnographic framework — AEIOU observations

Observing the Study Environment

Using the AEIOU ethnographic framework across student bedrooms, shared libraries, and home offices established clear physical and behavioural constraints for how an ambient air-quality device must live on a desk.

Code	Dimension	Observed Behaviours & Physical Context	Design Requirement
A	**Activities**	Sustained reading, essay writing, CAD modelling, and coding for 2–4 hours at a stretch.	Alert must be perceived effortlessly in peripheral vision without breaking flow.
E	**Environment**	Small single bedrooms and library cubicles with windows closed against cold or street noise.	Must sample local breathing-zone air directly on the desk surface.
I	**Interactions**	Deep screen focus punctuated by passive glances around the desk when mentally tired.	Gradual colour/pulse transition rather than an abrupt alarm or buzzer.
O	**Objects**	Laptop, monitor, notebooks, mug, desk lamp, and phone charger competing for space.	Compact Ø 60mm footprint powered via standard USB-C pass-through.
U	**Users**	University students, remote workers, and gamers seeking sustained mental clarity.	Warm botanical aesthetic that feels like a desk plant rather than lab gear.

System architecture — inputs & outputs map

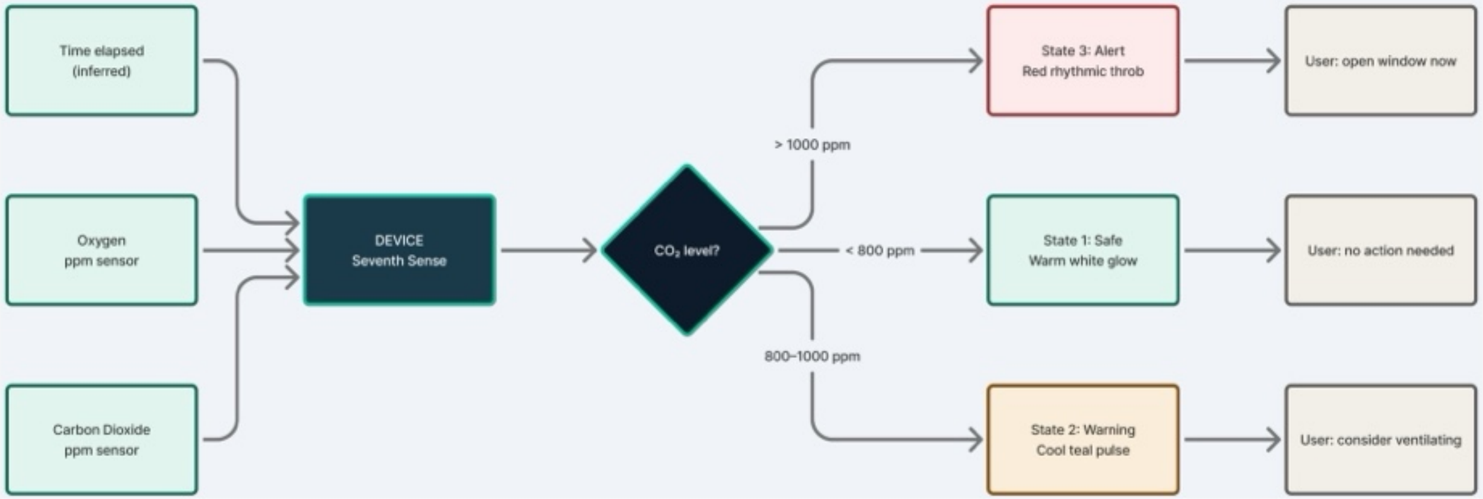


Fig. 07 — Inputs & Outputs system map evaluating environmental sensors vs human sensory outputs

Selecting the Optimal Input–Output Pair

Mapping all viable environmental inputs (CO_2, VOCs, temperature, humidity, noise) against sensory outputs (light, heat, scent, motion, haptics) clarified why photoacoustic CO_2 sensing paired with diffused botanical light offered the highest signal reliability.

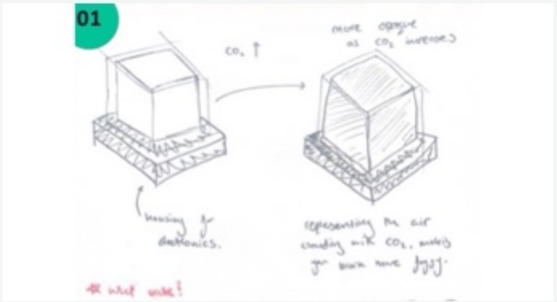
01 **Input: Photoacoustic NDIR CO_2** — Directly measures ppm concentration (400–2500 ppm) rather than estimating eCO_2 from VOCs.

02 **Processing: Threshold State Machine** — Smoothly interpolates NeoPixel colour and pulse speed across three clinical CO_2 bands.

03 **Output: Diffused Stigma Glow** — Fibre-like translucent pollen tubes channel light upward from the base into peripheral sight.

Initial ideation — seven sensory concepts

To translate invisible indoor CO_2 concentration into an intuitive 'Seventh Sense' without screens or companion apps, I sketched seven ambient product concepts spanning light, heat, scent, and magnetic resistance.



01 — Frosted Cube

Glass housing clouds more opaque as CO_2 rises, mirroring brain fog.



02 — Heat Rose

Emits rising warmth through a diffuser pod beneath the rose petals.



03 — Lily Light (Chosen)

Stigma & pollen tubes shift colour and brightness with CO_2 ppm.



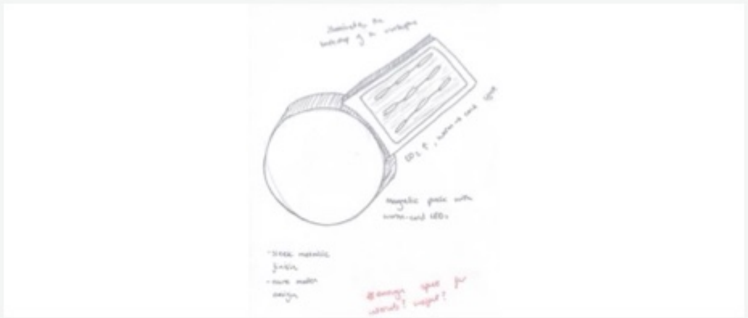
04 — Scent Diffuser

Releases stronger botanical scent as indoor CO_2 increases.



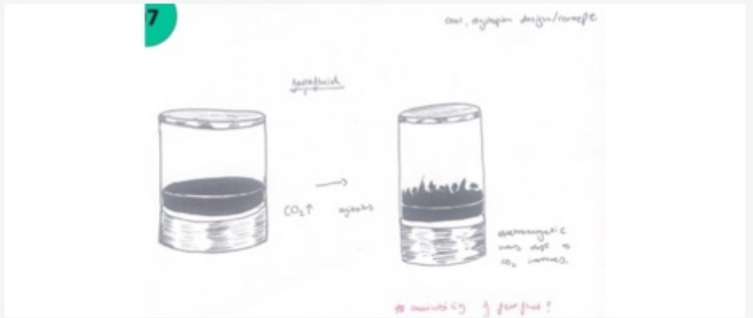
05 — Magnetic Fidget Bowl

Electromagnet strengthens as CO_2 rises, making steel ball harder to roll.



06 — Laptop Light Puck

Rear-mount magnetic LED shifts warm to cold across workspace backdrop.



07 — Ferrofluid Jar

Electromagnetic base agitates ferrofluid as air quality deteriorates.

Strongest alternate 1:1 sketch models



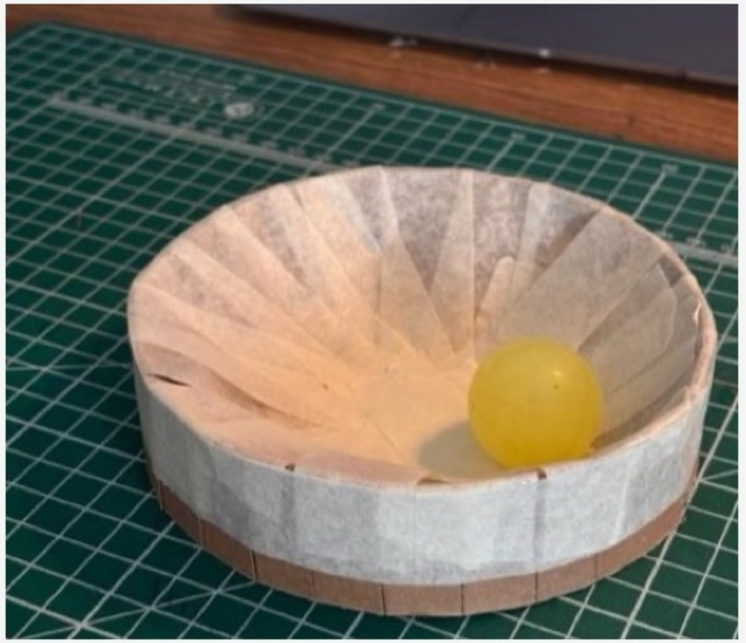
Model 01 — Heat Rose

Rounded cardboard base with perforated air vents beneath a real rose stem to test 1:1 desk presence and thermal diffusion ergonomics.



Model 02 — Laptop Light Puck

Thin magnetic puck mounted to the rear of a laptop lid with an LED array shifting warm-to-cool against the wall behind the screen.



Model 03 — Magnetic Fidget Bowl

Tensioned masking-tape parabolic bowl scaled around a weighted sphere to test tactile rolling resistance on a study desk.

Fig. 15–17 — Three 1:1 physical sketch models built to evaluate desk footprint, peripheral visibility, and sensory clarity

Chosen concept — The CO_2 Lily & three light states

Why the Botanical Lily Form Succeeded

Evaluating the 1:1 sketch models showed that a self-contained botanical object sitting quietly in peripheral vision communicated air quality far more intuitively than a hidden laptop puck or an active fidget toy.

01 **Natural Circular Housing** — Pot base naturally accommodates the 37mm circular Adafruit NeoPixel ring and SCD40 sensor.

02 **Peripheral Stigma Glow** — The central stigma and pollen tubes provide a distinct, non-glaring focal point for colour shifts.

03 **Minimal USB-C Footprint** — Ø 60mm base occupies less space than a coffee mug and runs continuously off desk USB-C power.

04 **Biophilic Metaphor** — Directly links a living plant form with the invisible oxygen/CO_2 respiration of the room.

State	Band	CO_2 Concentration	Light Behaviour	User Action
01	**Safe**	< 800 ppm	Calm warm white glow	Room is healthy; no action needed
02	**Warning**	800 – 1000 ppm	Cool teal gentle pulse	Air quality falling; ventilate soon
03	**Alert**	> 1000 ppm	Slow red rhythmic throb	Open window now; cognitive risk

Pre-Conscious Peripheral Cue

Rather than demanding attention when air is clean, State 1 blends into warm desk lighting. Only when CO_2 crosses 800 ppm does the shift to cool teal and rhythmic red gently draw the eye.

Technology investigation — hardware architecture









Fig. 18–21 — Internal hardware components selected for compact Ø 60mm base packaging

01 **Sensirion SCD40 CO_2 Sensor** — Miniature photoacoustic NDIR sensor (10.1 × 10.1 × 6.5 mm) measuring true CO_2 from 400–2,000+ ppm via I^2C.

02 **Seeeduino XIAO SAMD21** — Ultra-compact microcontroller (20 × 17.5 mm) with native USB-C power/programming and 3.3V logic.

03 **Adafruit 12-LED NeoPixel Ring** — 37mm outer diameter RGBW addressable LED ring positioned beneath the translucent stigma bundle.

04 **3D-Printed PLA & Translucent TPU** — Matte green/terracotta PLA structural shell paired with flexible translucent TPU stem and light-pipe tubes.

Internal block model — SolidWorks packaging

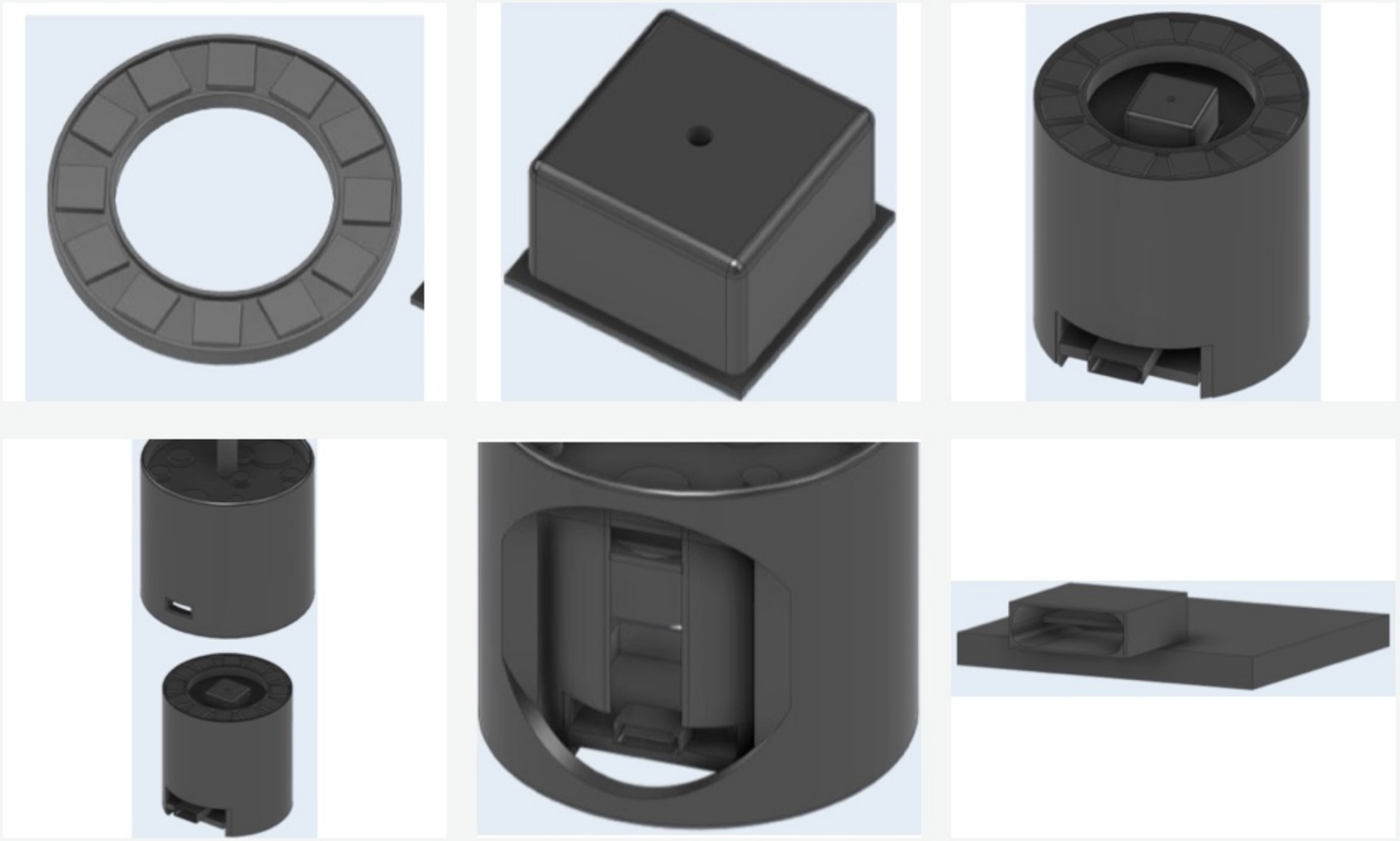


Fig. 22–27 — SolidWorks internal block modelling: verifying PCB clearance, NeoPixel shelf height, and USB-C port cutout

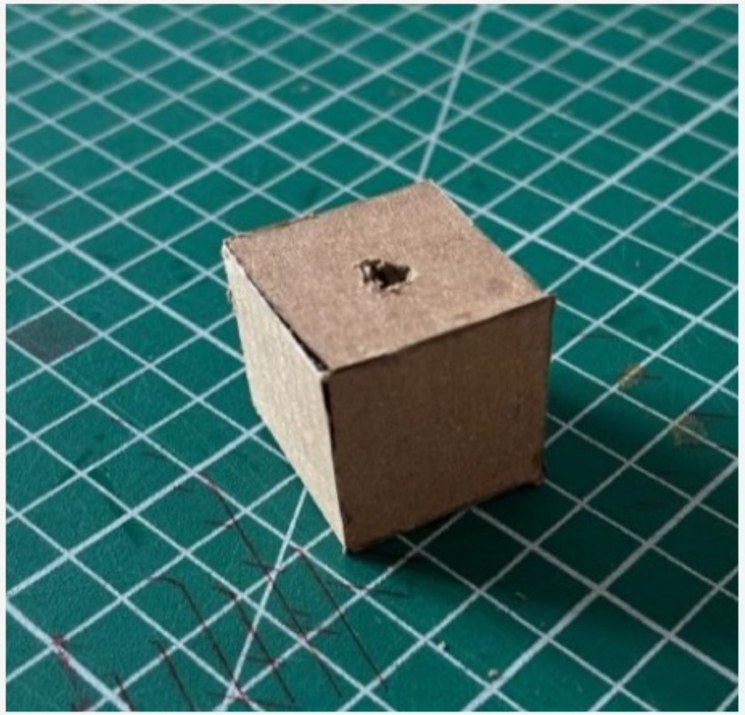
Volumetric Packaging in SolidWorks

Before sculpting the organic exterior in Fusion 360, I modelled exact dimensional blocks for the SCD40 sensor, Seeeduino XIAO, and 37mm NeoPixel ring in SolidWorks to establish the minimum viable pot diameter (60mm) and height (46mm).

01 **Thermal Isolation** — Separates the SCD40 sensing chamber from LED heat so readings stay accurate.

02 **Snap-Fit Lid Tolerance** — Tested annular lip clearances to allow tool-free access to the electronics.

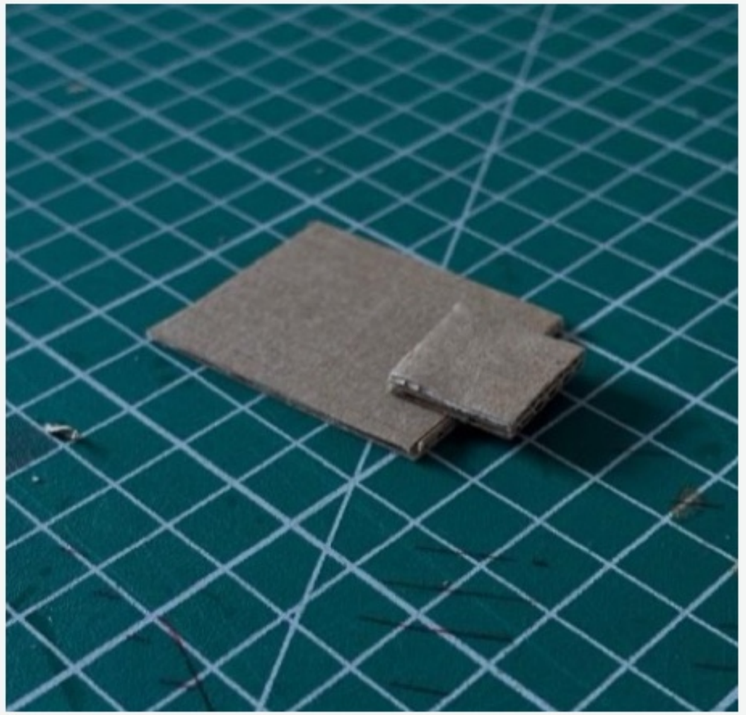
Component 1:1 physical sketch modelling



01 — Sensirion SCD40 Block (1:1)
Cardboard scale dummy verifying breathability vent alignment at the base of the pot.

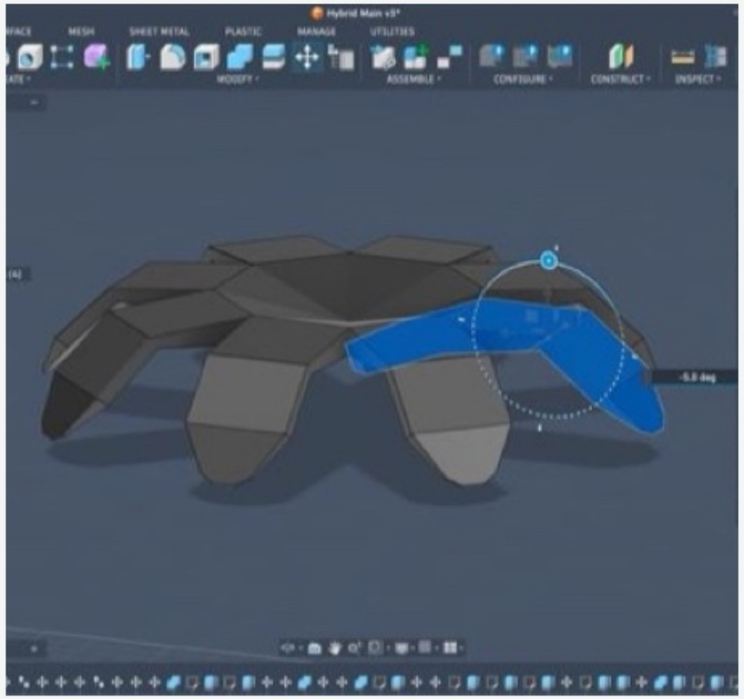


02 — 37mm NeoPixel Ring (1:1)
Physical ring template used to test upward light-pipe alignment into the lily stem.



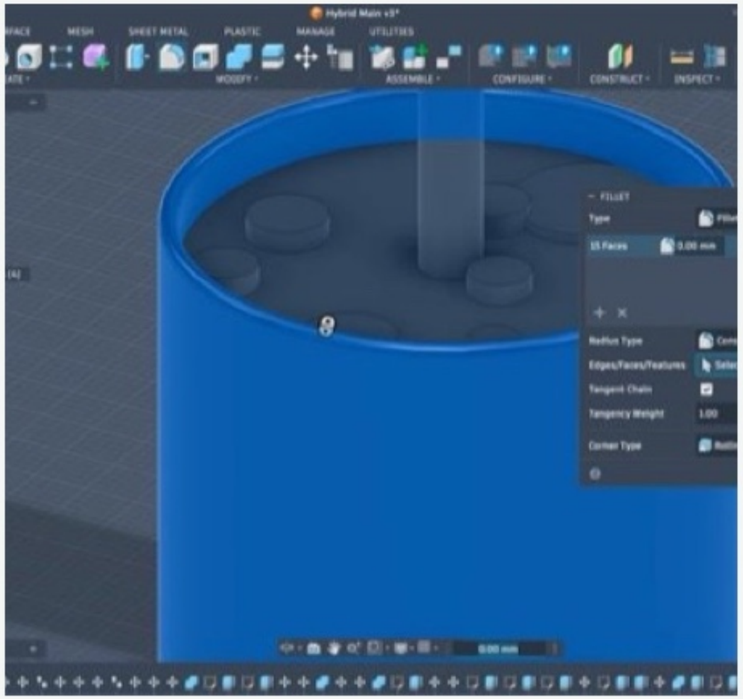
03 — Seeeduino XIAO + USB-C (1:1)
Confirmed rear USB-C cable strain relief and board seating on the internal base pillars.

CAD development — Fusion 360 organic surfacing



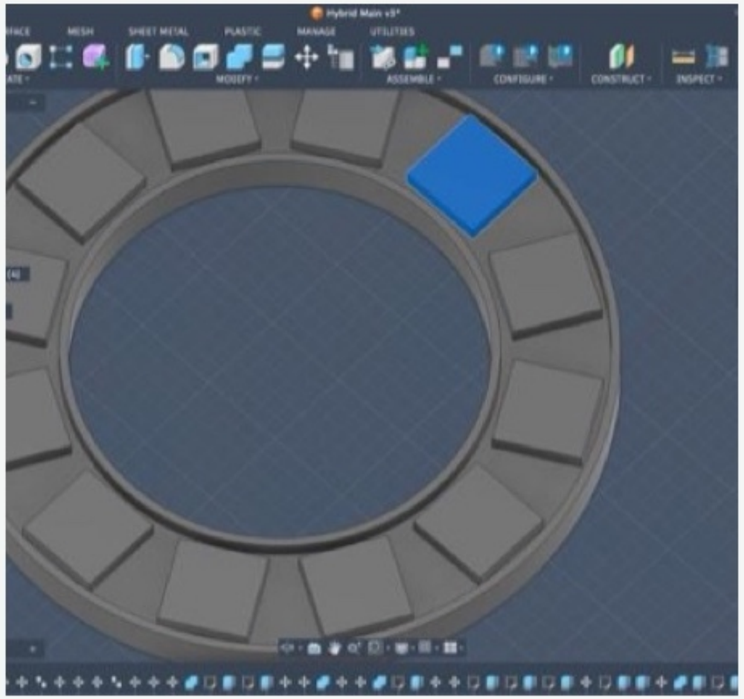
01 — Pot Base & Internal Pillars

Houses the microcontroller and sensor with four perimeter ventilation slots and a hollow central wiring column.



02 — Leaf Collar & Stem Conduit

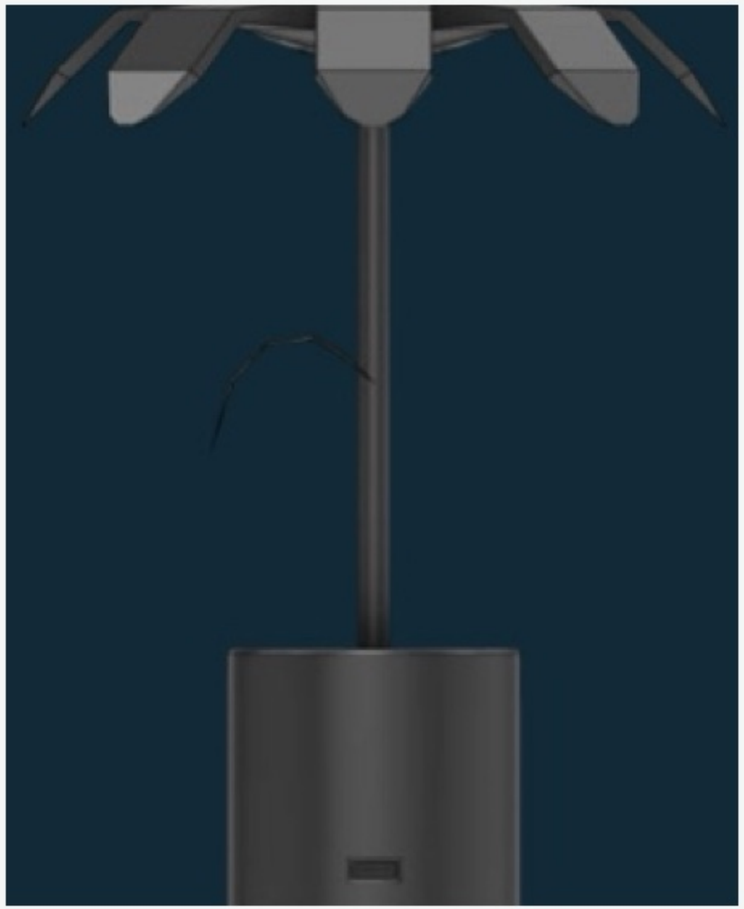
Curved botanical leaves conceal the pot lid seam while supporting the hollow TPU stem.

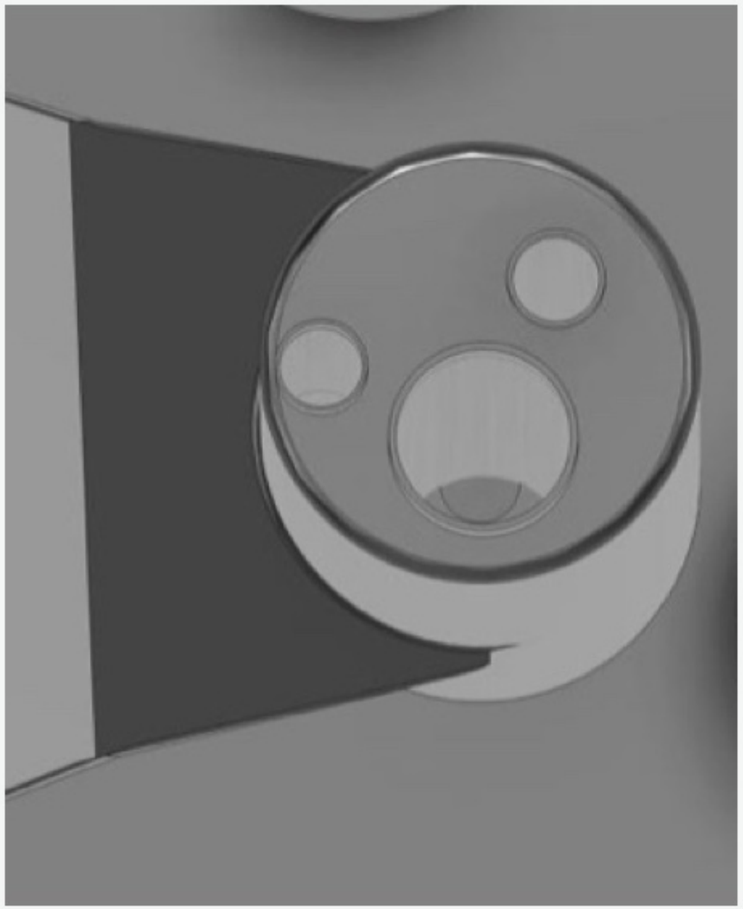


03 — Six-Petal Lily Head & Stigma

Lofted lily petals cradle six translucent pollen tubes and a central stigma that diffuse the NeoPixel glow.

Final design — multi-view CAD elevations





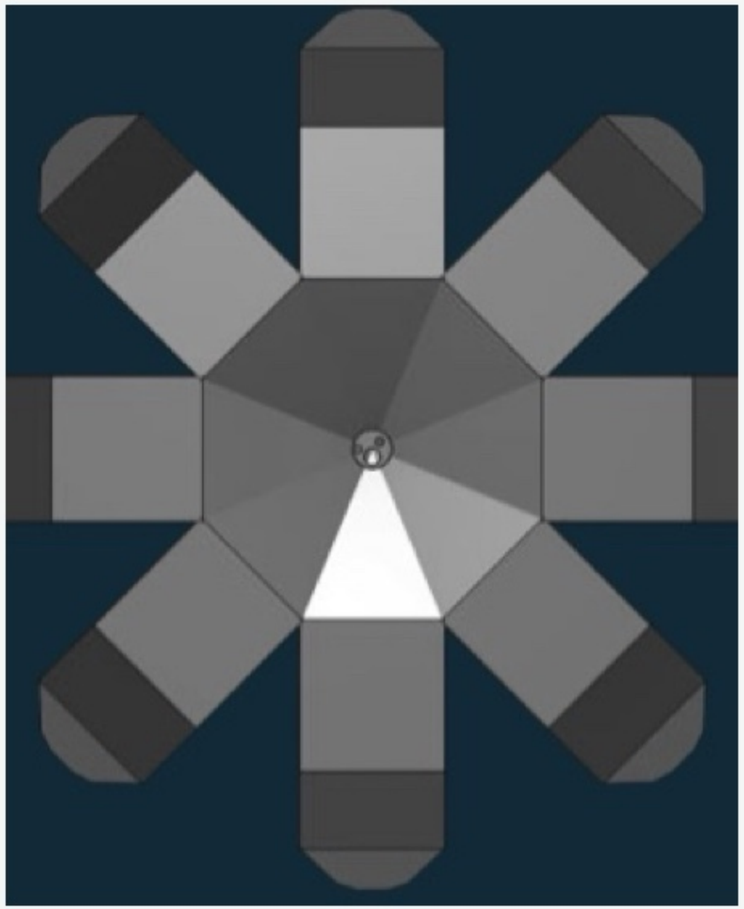
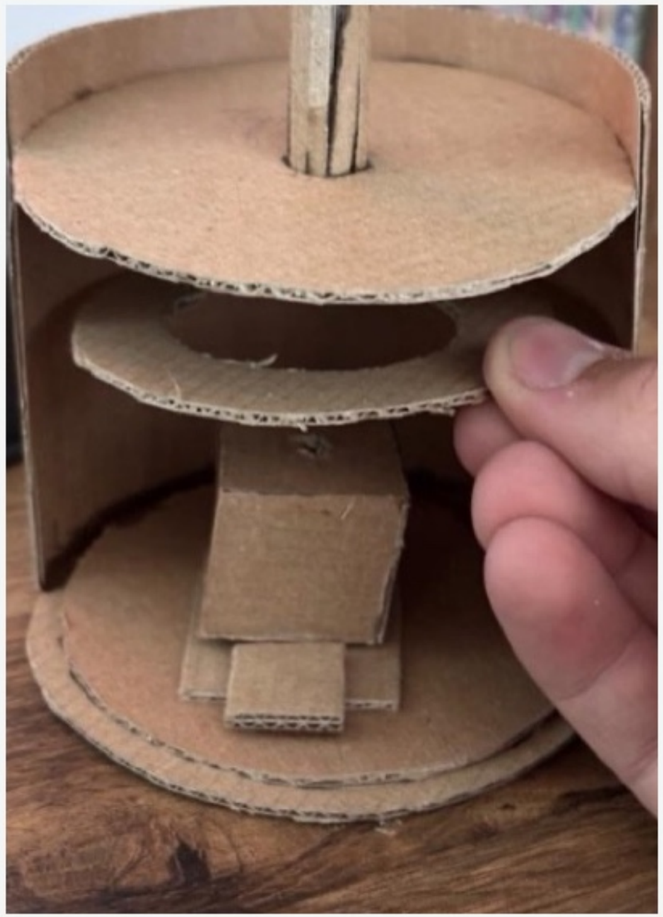


Fig. 31–33 — Fusion 360 final renders: Front elevation (165mm height), Internal stigma light-tube view, and Top radial petal array (Ø 95mm)

Final 1:1 soft model verification





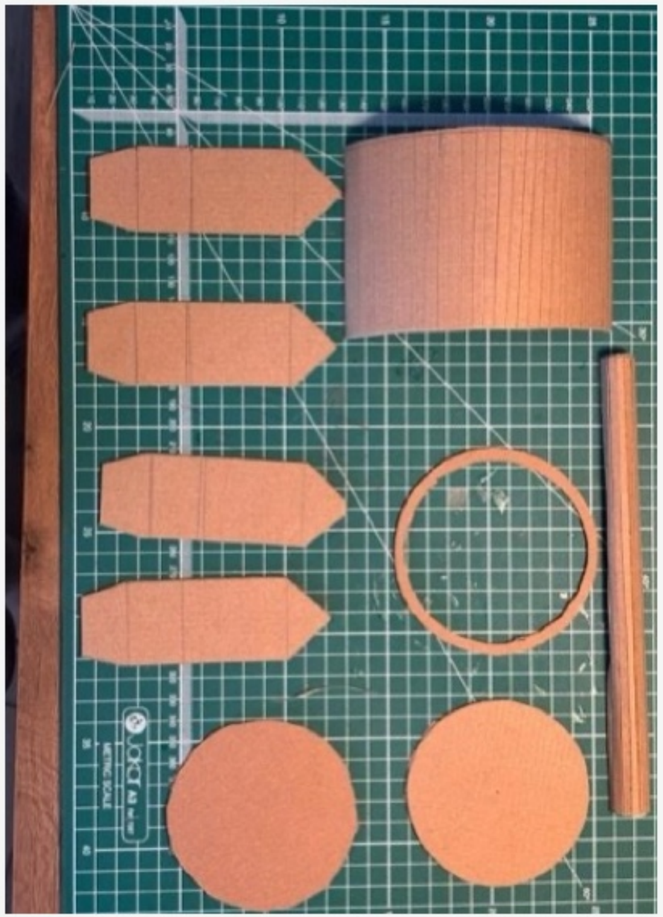


Fig. 34–36 — Full-size 1:1 cardboard and dowel soft model: testing desk proportion next to a laptop, internal sensor baffle stack, and petal curvature

User journey storyboard



01 — Student sits down (0 mins)

Fresh indoor air (<800 ppm) — Glowing Lily emits a calm, warm white glow.



02 — 45 mins into study session

CO_2 rising (800–1000 ppm) — cool teal pulse gently replaces warm white.



03 — 1 hr 30 mins (1,000+ ppm threshold)

Hypercapnia threshold reached — stigma transitions to a slow red rhythmic throb.



04 — Peripheral awareness

Peripheral red glow catches student's attention without a jarring screen pop-up.



05 — Window opened to ventilate

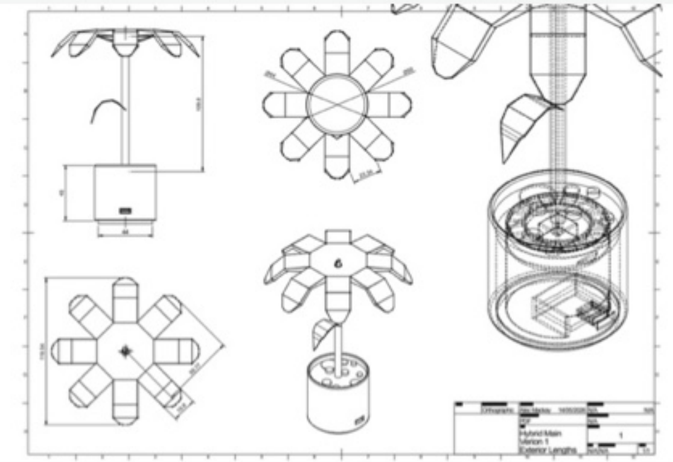
Student opens window; fresh outdoor air enters and CO_2 ppm drops.



06 — Room resets to <800 ppm

Warm white glow returns; focused study continues without cognitive crash.

Creative technologies pack — BOM & drawings





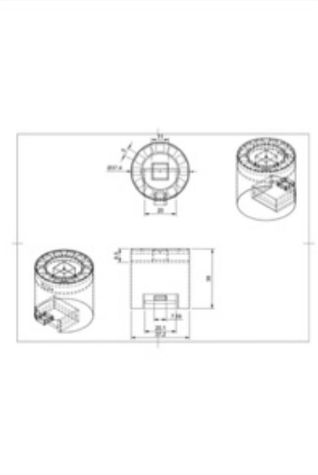


Fig. 43–45 — Exploded 12-part assembly, section view, and orthographic elevations (165mm × Ø95mm)

No.	Part Name	Qty	Material / Spec	Manufacturing Process
1	**Pot Base**	1	Matte Terracotta PLA	FDM 3D Print
2	**Pot Lid**	1	Matte Terracotta PLA	FDM 3D Print (snap-fit lip)
3	**Lower Leaf**	4	Matte Green PLA	FDM 3D Print
4	**Stem**	1	Flexible Green TPU	FDM 3D Print (hollow conduit)
5	**Lily Head**	1	Matte Cream PLA	FDM 3D Print
6	**Petal**	6	Matte Cream PLA	FDM 3D Print
7	**Pollen Tube**	6	Translucent Clear TPU	SLA / FDM Light Pipe
8	**Stigma**	1	Translucent Yellow TPU	SLA / FDM Diffuser Cap
9	**Pollen**	6	Matte Orange PLA	FDM 3D Print
10	**Microcontroller**	1	Seeeduino XIAO SAMD21	Off-the-shelf PCB (USB-C)
11	**CO_2 Sensor**	1	Sensirion SCD40	Off-the-shelf NDIR I^2C module
12	**LED Ring**	1	Adafruit 12-LED NeoPixel	Off-the-shelf Ø37mm RGBW PCB